CommerceFlow Test Strategy

Version 1.0 | 29 October 2026 | DESIGN DRAFT

1. Purpose

Verify requirements, secure user data, prevent overselling and demonstrate repeatable engineering evidence. No feature is considered done without risk-appropriate tests.

2. Test Levels

Unit: JUnit 5/Mockito and frontend component tests. Integration: Spring Boot Test with PostgreSQL Testcontainers and Flyway. E2E: Playwright critical journeys. Security: session, CSRF, RBAC, ownership. Performance: reproducible k6 workload.

3. Priority Scenarios

Registration duplicates and invalid inputs; wrong password; logout session invalidation; customer forbidden from admin; cross-customer order hidden; missing CSRF rejected; inventory concurrency; rollback; mock payment failure; idempotent retry.

4. Test Environments

Local Docker development; CI GitHub Actions with Docker-enabled integration tests; staging if provisioned; production only safe smoke tests. Never use real customer data or secrets in fixtures.

5. Definition of Done

Acceptance criteria satisfied; automated tests added and reviewed; CI passes; security behavior verified; documentation and requirement-to-test traceability updated; human review completed.

6. Evidence and Defects

Capture test file path, run URL, result, execution date and defects. Coverage is supporting evidence, not a substitute for meaningful assertions.